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A Palaeomagnetic Study of the Nigerian Volcanic Provinces

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Summary – Six Younger Granite localities showing normal and reverse magnetizations in equal proportion have given a Jurassic palaeomagnetic pole position $\phi = 62.5°$ N, $\Lambda = 241.6°$ E; (Fisher's precision parameter $(k) = 27.8$ and $\alpha_{95} = 13°$). Individual palaeopole-positions have also been obtained for a Cretaceous pyroclastic rock and for two Pleistocene basalt flows.

Introduction

There are few palaeomagnetic results from Africa over the span of Mesozoic to recent times and this study was planned to give an account of the palaeomagnetism of the Nigerian volcanic provinces that range over this period.

The sampling was made accordingly but in the course of the palaeomagnetic measurements more than two-thirds of the localities sampled turned out to be lacking primary magnetization. Also quite a number of samples of anomalous magnetization belonging to localities which are thought to have preserved original magnetization had to be rejected thereby lowering the accuracy of the estimation of the related site mean direction of magnetization.

Geology

In Nigeria there are three igneous provinces of post-Palaeozoic age. These are the Jurassic Younger Granites and associated rocks, the Cretaceous intrusions and finally the products of Tertiary to Recent volcanic activity.

The Younger Granites and associated rocks occur in a 250 km wide zone which extends northward from the Benue valley for 350 km. The main centre of igneous activity is the Jos plateau area. Minor occurrences of mainly volcanic rocks are also known near Daura at the Nigeria-Niger border (TURNER, 1971; TURNER and WEBB, 1974).

The Cretaceous igneous activity produced intrusions of basic and intermediate

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composition and also extrusive rocks which appear over the area of the Abakaliki anticlinorium in Eastern Nigeria (REYMENT, 1965).

The Cenozoic volcanics have a wide distribution in Eastern and North-Eastern Nigeria including the Jos plateau area. They form large basalt lava plateaux (Biu and Longuda plateaux), basalt and trachite – phonolite plugs, lava flows of basalt and cinder cones (Benue valley, Jos plateau) (GRANT *et al.*, 1972).

Radiometric dating has given values of around 160 m.y., a mid-Jurassic age, for the Younger Granites (JACOBSON *et al.*, 1963). The Cretaceous intrusions are of post-Albian age (REYMENT, 1965). Recent K/Ar age determinations have revealed two different periods of volcanic activity in the Cenozoic: the earlier is phonolitic-trachitic and is dated 22–11 m.y. ago and the later more widespread and chiefly basaltic, began 7 m.y. ago and has continued until recent times (GRANT *et al.*, 1972).

Methods applied

The sampling for this study was made by collecting oriented hand samples at an average rate of five per outcrop, mainly from natural outcrops.

From the 33 sampling localities which are listed with relevant data in Table 1 those numbered N1, N2,..., N24 with the exception of N5 and N16 represent Younger Granite occurrences. N5, N16 and N25–N31 are localities in the youngest volcanics, while N32 and N33 refer to two Cretaceous outcrops. Within each locality samples are numbered by additional numerals separated by a hyphen from the symbol of locality, e.g. N4-2 means the second sample of locality N4.

A.C. demagnetization technique was applied and the measurements of magnetization were carried out with a high sensitivity JR-2 (Jelinek) spinner-, and a low sensitivity home-made astatic magnetometer, respectively. Data processing was made on a RAZDAN-3 computer. Both measurements and data processing were executed in Hungary using the instruments and facilities of the Department of Geophysics, Eötvös University, Budapest.

Description of results

Younger Granite localities

Locality N1 exhibited poorly stable behaviour and large secondary magnetizations were removed by A.C. demagnetization. Optimum grouping was achieved at 160 Oe when all samples (with the exception of N4-4 which showed completely unstable behaviour) give a combined mean $\bar{D} = 349.2°$, $\bar{I} = -15.7°$ with $k = 15.8$ and $\alpha_{95} = 21.5°$.

Locality N2: Total NRM's are random but on A.C. demagnetization N2-1 shows

Table 1
Sampling localities.

Code of locality	Number of samples collected	Geographic		Rock type occurrence form
		Latitude°	Longitude°	
N1	5	8.82	8.51	Microsyenite, Dyke
N2	2	8.39	7.78	Basic Dyke
N3	3	8.71	8.52	Basic Dyke
N4	4	8.73	8.52	Basic Dyke
N5	6	7.51	8.59	Basalt, Intrusion
N6	6	10.77	8.71	Ignimbrite (Comendite)
N7	4	10.77	8.71	Rhyolite
N8	6	10.75	8.68	Basalt
N9	6	10.89	8.46	Rhyolite
N10	4	10.81	8.55	Granite-Porphyry, Ring-Dyke
N11	6	10.75	8.68	Rhyolite
N12	7(+3)	10.00	8.89	Dolerite Dyke
N13	6	10.03	8.87	Rhyolite
N14	6	10.07	8.88	Rhyolite
N15	7	9.53	8.92	Rhyolite
N16	10	9.37	8.99	Trachite, Basalt
N17	6	9.20	8.81	Rhyolite
N18	4	9.20	8.84	Rhyolite
N19	6	10.25	8.83	Syenite
N20	6	11.06	9.53	Rhyolite, Basalt
N21	3	11.06	9.53	Trachite
N22	7	11.06	9.53	Basalt, Rhyolite
N23	6	12.92	8.54	Syenite (Trachite)
N24	6	13.02	8.52	Rhyolite, Dyke
N25	6	9.78	11.25	Phonolite, Plug
N26	4	9.83	11.22	Basalt, Plug
N27	6	9.72	11.47	Phonolite, Plug
N28	4	9.31	12.23	Basalt, Plug
N29	4	10.39	12.54	Basalt, Flow
N30	4	10.60	12.13	Basalt, Flow
N31	6	10.24	11.47	Basalt
N32	4	6.35	8.09	Pyroclastic Rock
N33	3	5.90	7.84	Dolerite, Dyke

stable behaviour with a direction of magnetization which is approached by N2-2 after demagnetization to 400 Oe. The site mean is $\bar{D} = 343.8°$, $\bar{I} = +25.5°$.

Locality N3: Total NRM's exhibited a strung distribution but on A.C. demagnetization N3-1 showed a fairly stable behaviour with a direction of magnetization at 160 Oe which is approached by N3-3 after demagnetization to 400 Oe. (While N3-2 also showed a weak tendency to group N3-4 was completely unstable.) The site mean is $\bar{D} = 139.3°$, $\bar{I} = +26.0°$.

Locality N12: Total NRM's showed a strung distribution which, though significantly reduced by progressive A.C. demagnetization, remained characteristic of the

magnetization. N12-1 however showed a stable direction of magnetization around which those of N12-2, N12-3 and 12-5 clustered after demagnetization of 400 Oe. The combined mean is $\bar{D} = 153°$, $\bar{I} = +31$ with $k = 20$ and $\alpha_{95} = 28.3°$.

Locality N15: Best grouping was achieved by piecing together the directions of magnetization after demagnetization to 250 Oe for N15-2 and N15-6 and to 400 Oe for N15-1, N15-3 and N15-7, which gave a site mean $\bar{D} = 334.6°$, $\bar{I} = -14.4°$ with $k = 50$ and $\alpha_{95} = 11°$.

Locality N17: Three samples N17-3, N17-4 and N17-6 gave excellent grouping of the directions of magnetization after demagnetization to 400 Oe. The remaining samples N17-1, N17-2 and N17-5 showed only a weak tendency for their directions of magnetization to approach the site mean which is $\bar{D} = 166.6°$, $\bar{I} = +19.9°$ with $k = 500$ and $\alpha_{95} = 5.5°$.

Other Younger Granite localities. Meaningful palaeomagnetic directions were isolated from none of the samples collected from these localities, but there are differences in behaviour to progressive A.C. demagnetization which allow the samples to be assigned to three groups: (i) all individual directions of magnetization were fairly stable as in N10 and N11; (ii) the directions of magnetization were, with the exception of that of one sample or two, also stable as in N4, N9, N13, N14, N18, N20, N22, N23 and N24; (iii) major directional changes but with no reduction of dispersion took place almost in all samples, usually along different great circles as in N6, N7, N8, N12-(8–10), N19 and N21.

Cretaceous localities

Locality N32 exhibited stable behaviour on A.C. demagnetization and the site mean was computed from the total NRM to give $\bar{D} = 342.3°$, $\bar{I} = +1.4°$ with $k = 600$ and $\alpha_{95} = 3.7°$.

Locality N33: No grouped directions of magnetization were isolated from samples collected here.

Cenozoic localities

Locality N29 exhibited a large dispersion of the total NRM's but a fairly precise mean direction of magnetization was achieved after demagnetization to 400 Oe (250 Oe for N29-4), which is $\bar{D} = 172.7°$, $\bar{I} = -5.6°$ with $k = 200.0$ and $\alpha_{95} = 6.5°$.

Locality N30 gave the smallest dispersion of the directions of magnetization after demagnetization to 160 Oe. The site mean is $\bar{D} = 17.9°$, $\bar{I} = +24.6°$ with $k = 54.5$ and $\alpha_{95} = 12.5°$.

Locality N31: A strung distribution is characteristic of this group with a large scatter of the directions of magnetization at all except the 80 Oe stage of demagnetiza-

tion, where the mean direction of magnetization: $\bar{D} = 332.3°$, $\bar{I} = -13°$ has a precision $k = 38.5$ and, $\alpha_{95} = 11°$.

Other Cenozoic localities: Palaeomagnetic directions could be isolated from none of the samples here. All individual directions of magnetization were stable in N16 and completely unstable in N5 and N27 on A.C. demagnetization. In the remaining groups (N25, N26, N28) both types of magnetization were displayed.

Ore microscopy

The results of microscopic observations made on 25 samples from 21 localities, N1–N22 (except N4) are presented in Table 2.

The interpretation of Table 2 for the samples used palaeomagnetically is quite obvious, their primary magnetic grains being unaltered. However, there are also samples with no microscopically visible sign of post-primary mineralogical alteration on the majority of the grains which are still without primary magnetization. For this either mineralogical alteration not seen under the microscope or secondary magnetizations masking the original or both may be responsible.

The presence of deep alteration of the magnetic grains in some samples can easily be attributed to surface weathering. Samples with such grains naturally show no primary magnetization.

Discussion

Jurassic: Site mean directions of magnetization have been determined for six localities (N1, N2, N3, N12, N15, N17). They may each represent the local direction of the ancient field. As the estimation of k the precision of the mean direction of magnetization is meaningless for a mean composed of less than 3 individual directions and also unreliable for few observations the accuracy of the site means under consideration cannot formally be judged. Therefore each site mean will be regarded as a single observation for the related field direction. These are pieced together by computing 'site poles' which will, irrespective of polarity, be averaged out to obtain a single significant palaeomagnetic pole position as follows: $\phi = 62.5°$ N, $\Lambda = 241.6°$ E with $k = 27.8$ and $\alpha_{95} = 13°$. Presumably, the effect of the secular variation and of possible dislocations of local nature are somewhat diminished because of averaging and the pole so defined does satisfy certain minimum reliability criteria among which are the presence of normal and reverse magnetizations (3–3 sites) and the divergence of the directions of magnetization from the present field direction.

A number of Jurassic pole positions for Africa have been obtained previously, e.g. IRVING (1964), CREER (1970). Creer's mean pole ($\phi = 68°$ N, $\Lambda = 256°$ E; $\alpha_{95} = 18°$) differs by an insignificant 8° of arc from our Nigerian pole.

Table 2
Microscopic observations.

Samples with primary opaque minerals		Samples with altered primary opaque minerals	Samples with other†) opaque minerals	
Predominant magnetite	Predominant ilmenite			
N1-1: magnetite up to 1000 μm size with ilmenite plus heamatite in exsolution; ilmenite of similar size with exsolved heamatite; subordinate pyrite. *N15-1:* as N17-6 below. *N17-6:* homogeneous magnetite (≦ 10 μm); also highly reflective opaque connected with mafic minerals.	*N2-1:* ilmenite (100 μm) with exsolved heamatite; ilmenite (10 μm); magnetite (10 μm). *N12-5:* ferric ilmenite (200 μm) with patches of typical ilmenite; magnetite (200 μm) with exsolved ilmenite; small (< 10 μm) homogeneous grains of magnetite. *N15-7:* no magnetite; ilmenite in three generations: larger grains (to 100 μm) evenly distributed in the matrix, small ones (15 μm) as exsolution in pyroxenes; subordinate pyrite, rutile and hydrous iron oxide.			Localities considered palaeomagnetically
N5-3: Large grains (~ 1000 μm) of magnetite and ilmenite often intergrown, both are homogeneous. *N7-3:* magnetite (30–50 μm) with ilmenite and heamatite in exsolution aligned in flow directions; magnetite (< 5 μm); ilmenite (100 μm) with exsolved heamatite. *N14-4:* as N7-3 with the exception of ilmenite which is homogeneous and < 30 μm long. *N16-9:* homogeneous magnetite (10–15 μm). *N22-4:* heterogeneous magnetite containing lamelliform and patchy heamatite in exsolution.	*N12-8:* homogeneous ilmenite (15–200 μm); magnetite(50–300μm) with exsolved ilmenite; subordinate limonite on part of mafic minerals. *N19-3:* ilmenite (100 μm) with exsolved heamatite, magnetite (10 μm) homogeneous, few shows incipient maghemitization, hydrous iron oxide on mafic minerals.	*N9-4:* magnetite (≦ 10 μm) with maghemite up to 70% of the grains. *N10-1:* magnetite with heamatite in exsolution (≦ 50 μm); ⅓ of the grains shows advanced maghemitization. *N11-4:* euhedral magnetite grains totally maghemitized; irregularly shaped heamatite and/or hydrous iron oxide (15–50 μm). *N16-3:* magnetite (≦ 20 μm) in various states of alteration producing maghemite and heamatite. *N18-3:* magnetite (≦ 50 μm) visible maghemitization on 20–40% of the grains. *N20-4:* limonite from primary ilmenite (20–50 μm).	*N3-2:* opaque minerals in aggregates, also small grains (< 8 μm) of strong reflexion; rutile. *N6-5:* fine grained (< 5 μm) highly oxidized anhedral opaque evenly distributed in the ground mass plus thin veins of iron hydroxide. *N8-6:* opaque minerals in aggregates *N13-4:* fine grained (≦ 15 μm) highly oxidized anhedral opaque of even distribution. *N21-3:* opaque minerals in aggregates. *N22-6:* irregular areas of highly reflective opaque plus as in N13-4	Localities not considered palaeomagnetically

†) Anhedral small grains cannot be specified closer.

Among the African poles in the Jurassic there is one for the Marangudzi and Mateke Hills Ring Complexes, Nuanetsi igneous province, Southern Rhodesia (GOUGH *et al.*, 1964) which naturally lends itself for comparison with the pole for the Nigerian Younger Granites because of the close relationship between the geological setting of the two provinces in question and their being approximately of the same age. Gough and co-workers give an overall mean pole position for 12 localities studied, which is $\phi = 60.0°$ N, $\Lambda = 270.2°$ E with $k = 31$ and $\alpha_{95} = 7.9°$ but prefer the following mean: $\phi = 60.9°$ N, $\Lambda = 266.2°$ E with $k = 57$ and $\alpha_{95} = 6.5°$ obtained for 10 sites of reverse magnetization only (GOUGH *et al.*, 1964).

The divergence of the Nigerian pole from these is 14° and 12° of arc, respectively. On Watson's *F*-test (WATSON, 1956) the Nigerian pole is not different from either of the Rhodesian results at the 95 percent level of confidence. At a confidence level between 90 and 95 percent however, while the precisions continue to be statistically identical, the Nigerian pole position becomes significantly different from both Rhodesian poles. This difference of pole positions could be interpreted in terms of tectonic movements but we feel this has to be postponed until more data are available.

Cretaceous: The only site mean derived here from the directions of magnetization of four specimens of two samples for locality N32 however precise ($k = 600$) must be taken merely as an isolated result which needs further corroboration.

The corresponding pole position is $\phi = 71.2°$ N, $\Lambda = 256.8°$ E ($\Delta p = 3.7°$, $\Delta m = 1.9°$). For comparison's sake Creer's Cretaceous pole for Africa (CREER, 1970) is $\phi = 71°$ N, $\Lambda = 221°$ E with $\alpha_{95} = 17°$.

Cenozoic: The basalt at locality N29 has been given a K/Ar age of 1.4 m.y. (GRANT *et al.*, 1972) and that at locality N30 is probably of around this age or younger. The magnetization of N29 is reverse and that of N30 is normal. The presence of both polarities may indicate stability. The two axes corresponding to the two poles ($\phi = 80°$ N, $\Lambda = 239.6°$ E ($\Delta p = 5.6°$, $\Delta m = 2.8°$) for N29 and $\phi = 72.7°$ N, $\Lambda = 90°$ E ($\Delta p = 13.4°$; $\Delta m = 7.2°$) for N30) are making an angle of 27° which may be the result of either secular variation or tectonic movements or both.

Palaeomagnetic poles of comparable age reported recently from Africa may be found in (PIPER and RICHARDSON, 1972) and in (REILLEY, 1970).

The Dadin Kowa Basalt at locality N31 has been given a K/Ar age of 84 m.y. (FREITH, 1970) which was later considered merely 'to be the result of excess 40Ar' (GRANT *et al.*, 1972). As for the magnetization of this basalt the A.C. demagnetization characteristics give evidence that it is the resultant of several soft magnetizations of probably viscous origin. The mean direction of magnetization at the 80 Oe stage of A.C. demagnetization recorded in the preceding section is regarded as that of one of these secondary magnetizations.

Finally, Table 3 sums up all relevant palaeomagnetic information which has been derived from the present study.

Table 3
Palaeomagnetic results.

Locality								Palaeo-magnetic polarity					Remark
Code	$\varphi°$	$\lambda°$	N/N_0	$\bar{D}°$	$\bar{I}°$	k	α_{95}		$\phi°$	$\Lambda°$	$\Delta p°$	$\Delta m°$	
						Jurassic							
N1	8.82	8.51	4/5	349.2	−15.7	15.8	21.5	+	69.8 N	221.1 E			
N2	8.39	7.78	2/2	343.8	+25.5	–	–	+	73.3 N	296.6 E			
N3	8.71	8.52	2/4	139.3	+26.0	–	–	–	43.7 S	69.7 E			
N12	10.00	8.89	3/7	153.0	+31.0	20.0	28.3	–	52.3 S	54.1 E			
N15	9.53	8.92	5/7	334.6	−14.4	50	11.0	+	59.7 N	240.2 E			
N17	9.20	8.81	3/6	166.6	+19.9	500	5.5	–	66.4 S	43.3 E			
							Average:	Mixed	62.5 N	241.6 E	$K = 27.8$	$\alpha_{95} = 13.0$	
						Cretaceous							
N32	6.35	8.09	2/4	342.3	+1.4	600*	3.7*	+	71.2 N	256.9 E	3.7	1.9	*computed from four specimens
						Pleistocene							
N29	10.39	12.54	4/4	172.7	−5.6	200	6.5	–	80.0 S	59.6 E	5.6	2.8	
N30	10.60	12.13	4/4	17.9	+24.6	54.5	12.5	+	72.2 N	90.0 E	13.4	7.2	

Legend: $\varphi°$, $\lambda°$ = geographic coordinates of locality; N/N_0 = total number of samples evaluated/collected; $\bar{D}°$ = site mean declination of magnetization; $\bar{I}°$ = site mean inclination of magnetization, + = downward; k = precision; α_{95} = semiangle of cone of 95% confidence for site mean direction (FISHER, 1953); $\phi°$, $\Lambda°$ = geographic coordinates of palaeomagnetic north-pole; $\Delta p°$, $\Delta m°$ = semiaxes of oval of confidence for palaeomagnetic pole position (FISHER, 1953).

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